Against Practical Science

(The Curious Case of the Missing Unicorns)

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Scientific Unicorns

Big ideas that propel human progress

(1) Rare
(2) Hard to appropriate

This talk

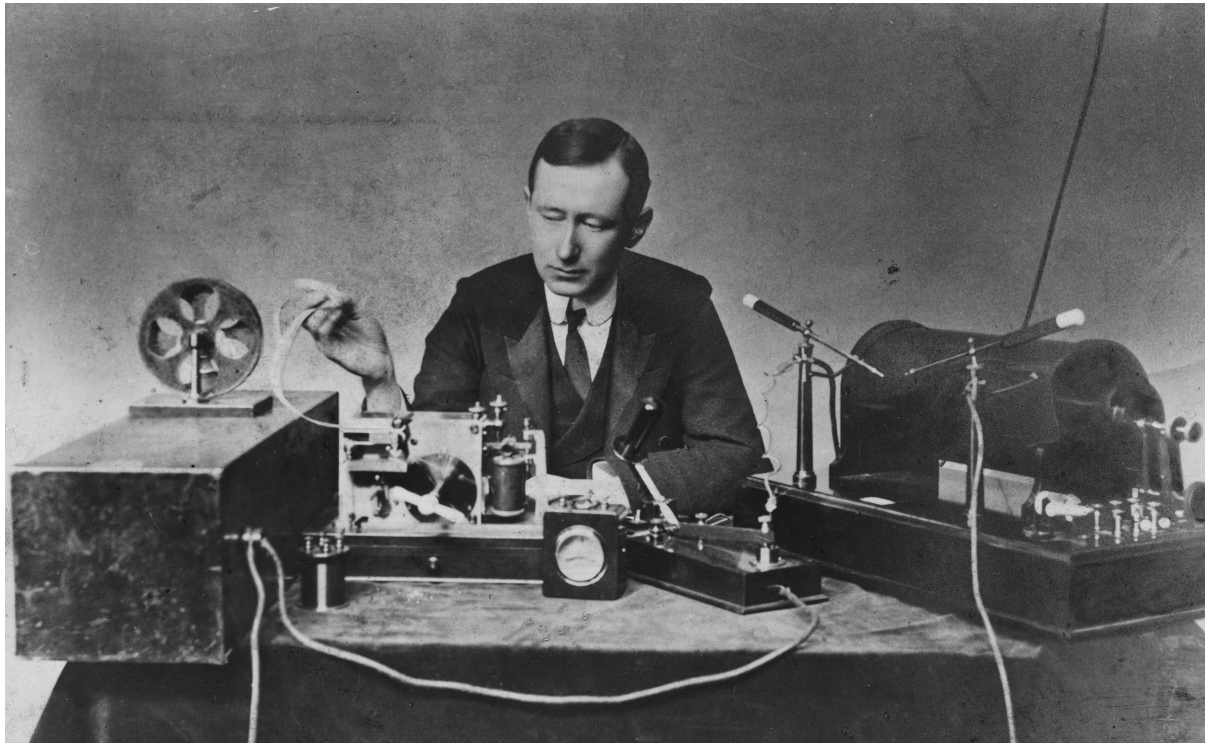
- Society wants unicorns
- But the way we operate is killing them
- We can do better

A Story

- **1860s:** James Clerk Maxwell
- **1880s:** Heinrich Hertz
  - *"It is of no use whatsoever... this is just an experiment that proves Maestro Maxwell was right."*
- **1890s:** Guglielmo Marconi
- **1897:** UK government
- **What technology?**

Radio

- The most "useful" technology of the 20th century started as a "useless" theory with "useless" experiments confirming it.





The Vending Machine Model

- Current Policy Premise:
  1. Need useful solution
  2. Input $
  3. Press Button
  4. Get Solution

By selecting on expected usefulness,

we *kill* unicorns

Three Problems:

1. Unborn Unicorns

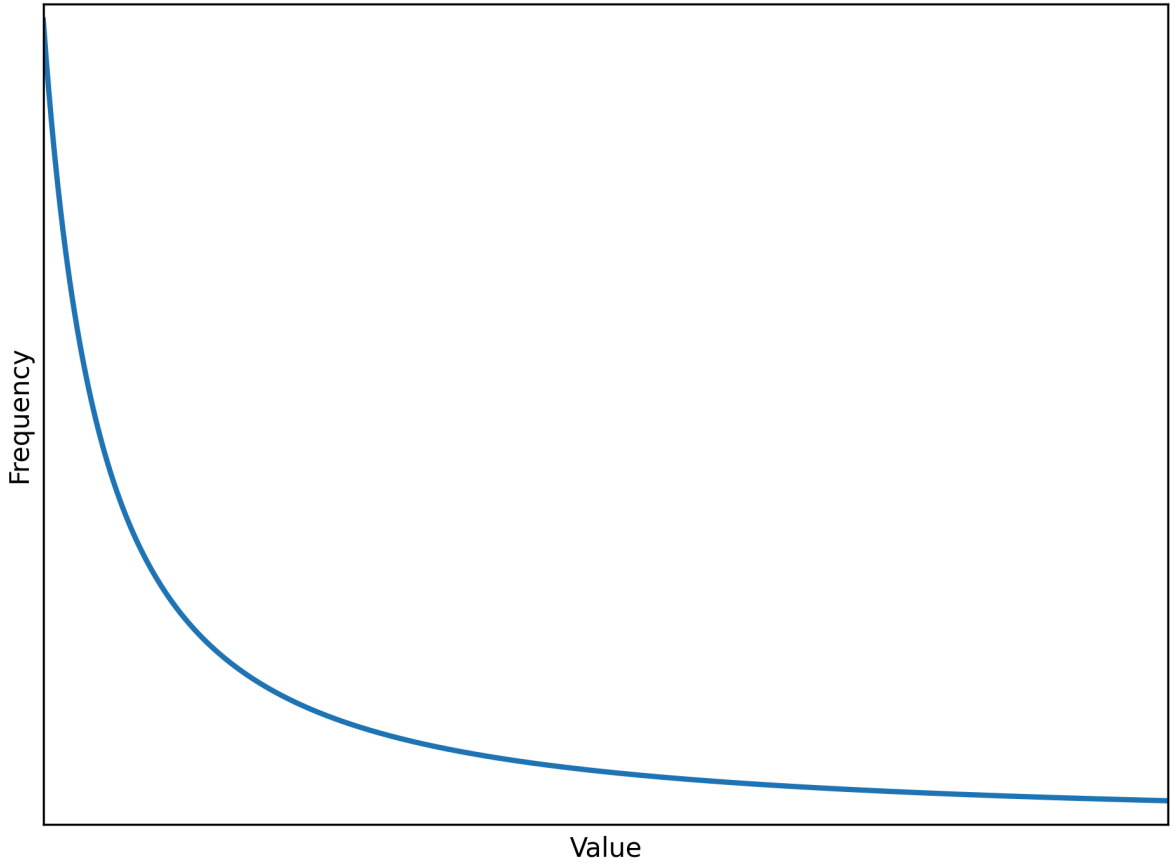
2. Future Unicorns

3. Gift Unicorns

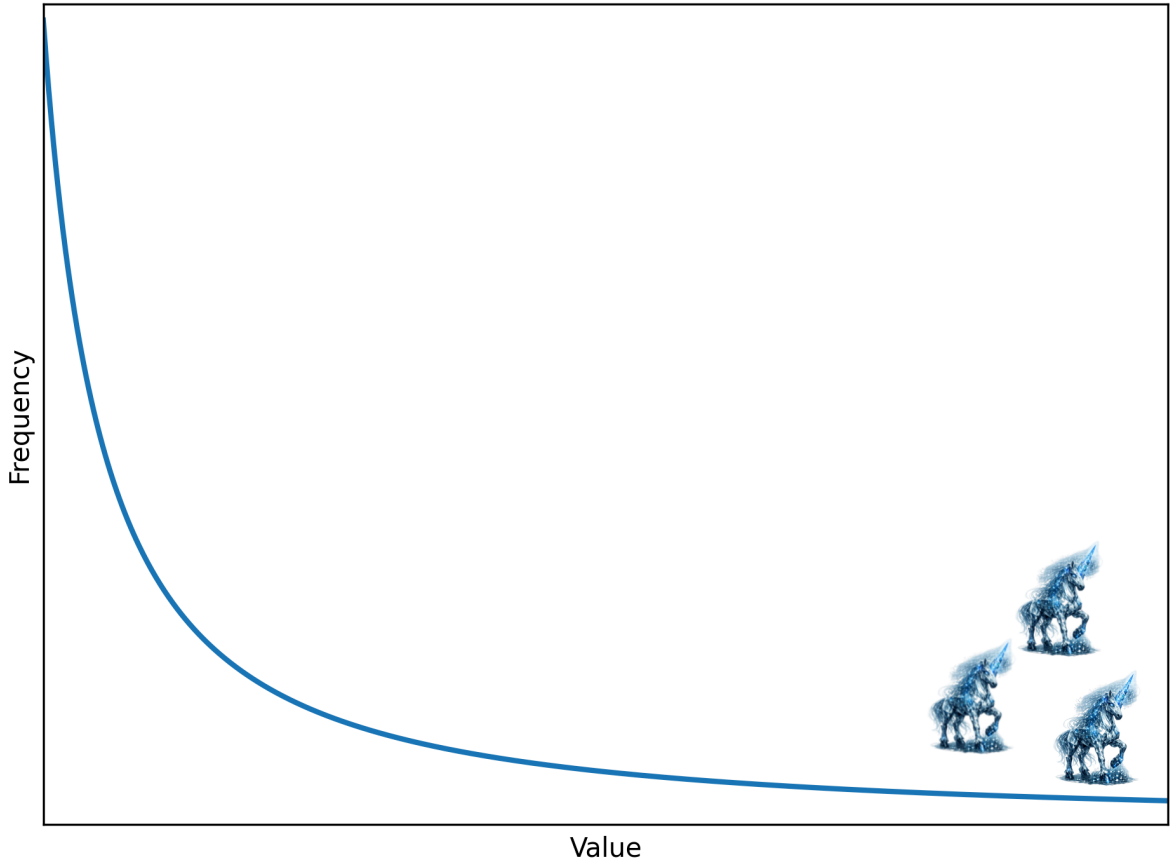
Problem 1: Unborn Unicorns

What is the value of scientific projects?

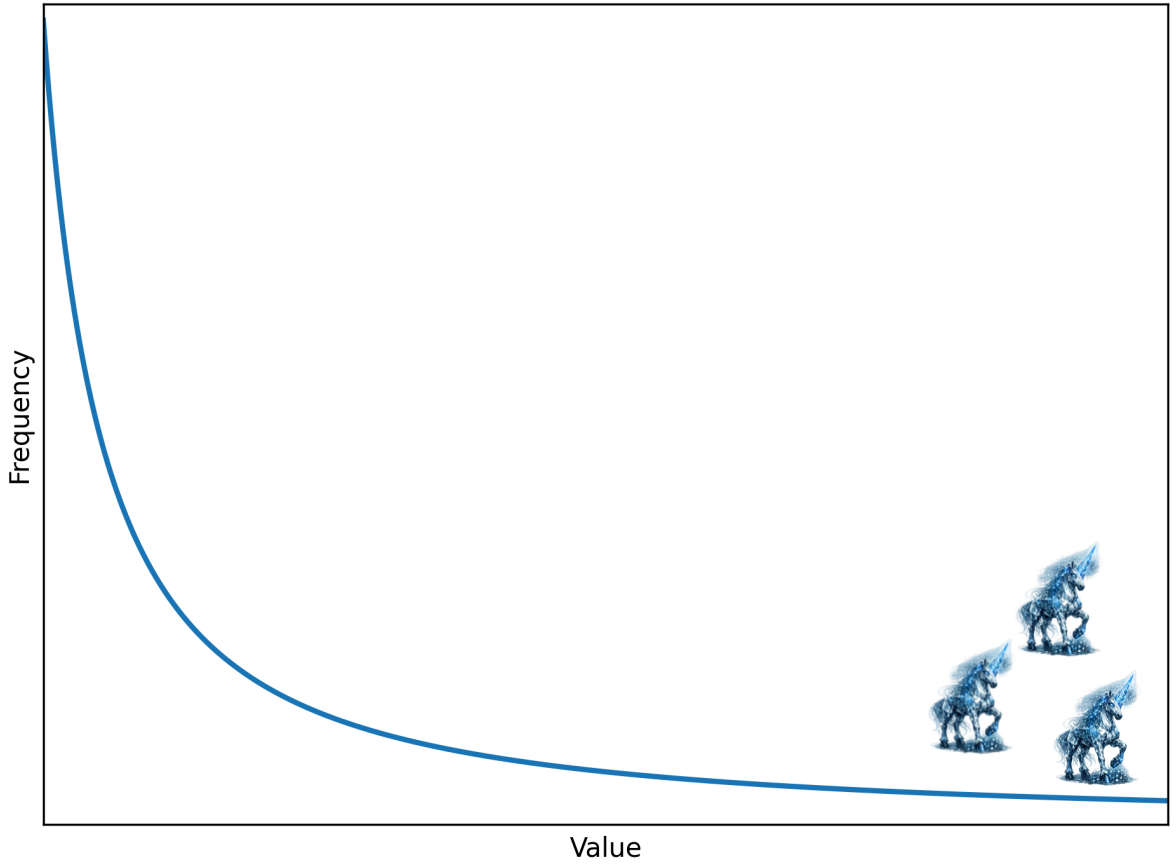
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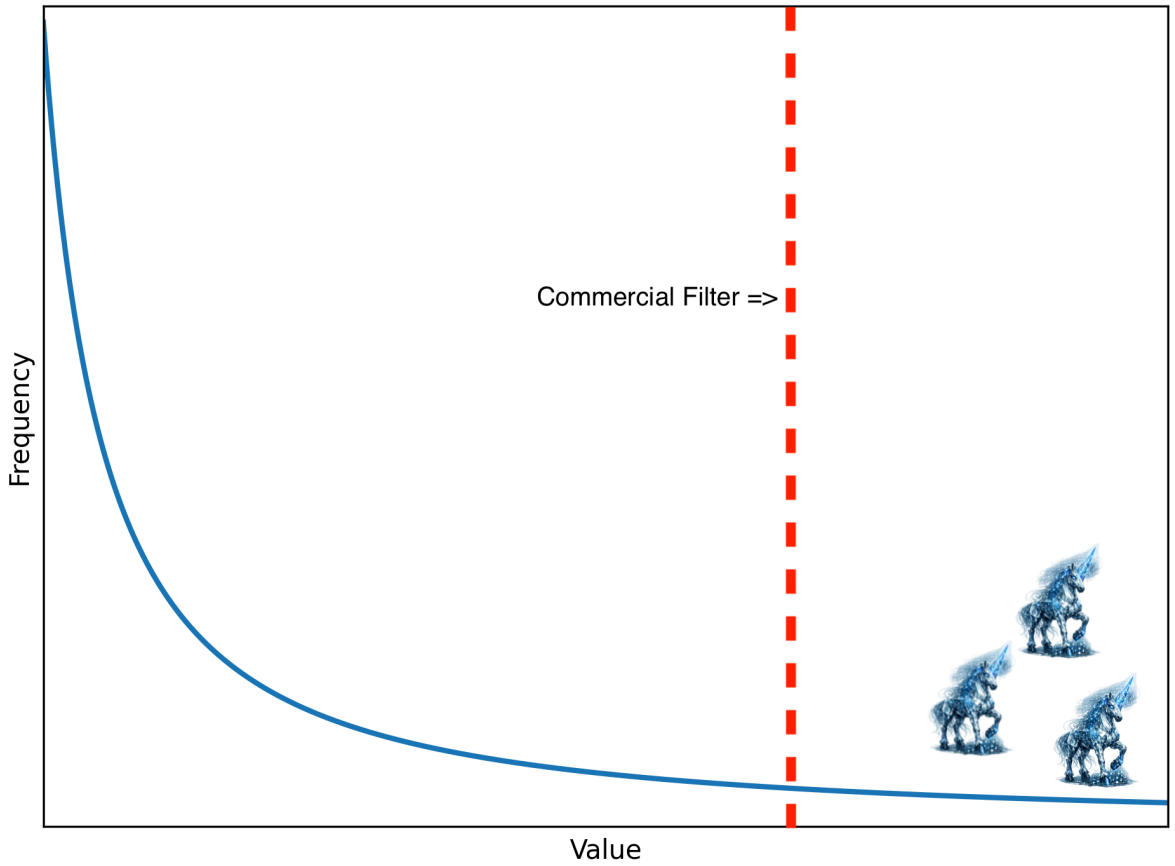
Unicorns: rare & hard to appropriate



Conventional Wisdom

"We need useful knowledge: let's focus on what can be commercializable / patentable / impactful"

Unicorns: cut off by the filter



But Wait!

"If it were valuable, the private sector would do it!"

This assumes that the private sector:

1. Is willing to take on lots of risk
2. Is more future-oriented than the rest of society
3. Doesn't care about appropriability

- All of these are false!

Deeper problem:

We are hopeless at picking winners

Innovation	Prediction	Reality
Laser (1960)	"A solution looking for a problem"	Surgery, telecom, manufacturing
Transistor (1947)	"Limited commercial applications"	Foundation of all computing
Talkies (1927)	"Who wants to hear actors talk?"	Ended silent film era

Deeper problem:

We are hopeless at picking winners (both ways!)

Innovation	Prediction	Reality
Google Glass (2013)	"Will change everything"	Discontinued
Segway (2001)	"Will transform cities"	Mall cop transport
Theranos (2014)	"The next Apple"	Fraud conviction

Problem 2: Future Unicorns

Science as insurance

- You are playing poker
- You can pay $5 to see the next card then decide next move
- If the next card is good => useful info!
- If the next card is bad => also useful info!
- The $5 payment is not wasted in any case

What should you fund?

Year 1: Start with 10 promising projects

→ Invest $1m in each = **$10m**

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Year 4: One last failure, but...
→ **The remaining project earns $100m** 🎯

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💰 **$30m invested → $100m returned = $70m profit**

Option Value of Science

- Instead of funding what (we think) works, we need to fund *options*
- **Not waste**: Useful information either way!
- Instead, we require impact statement / impact evaluation
- Must provide value within short timeframe

It gets worse

- If we use impact as a criterion for funding,
- we are chopping down the branches of the problem space...
- ... where unicorns live
- We are putting all our eggs in one/few basket(s)

Can we go back and get the unicorn?

- Knowledge is cumulative
- If we do not keep the branch where the unicorn is hiding,
- The branch withers and dies.
- Entire chunks of the problem space shrink
- Don't believe me - Let's look at Bell Labs

How Bell Labs Withered

- Credited with developing: transistor, laser, PV, Unix, C++, ...
- 11 Nobel Prizes; 5 Turing Awards

Until... Bell Labs failed an appropriability test

- Anti-trust regulation broke up Bell
- Bell Labs became a subsidiary of AT&T, i.e. a business unit
- AT&T asked Bell Labs to justify its existence
- It couldn't. Entire fields shrank (physics, acoustics, and more)

Problem 3: Gift Unicorns

(some of these get to live)

Unicorn Politics

A brief aside on Job-Ready Graduates

- Australian, but not uniquely
- Employers want job-ready graduates
- Universities train job-ready graduates
- On the face of it, fair enough
- Who *doesn't* want to be skilled at their job?

What's the Alternative?

- Instead of graduates with a set of **specific skills**,
- We train graduates with **general skills**: think / learn / evaluate / analyse / ...
- What do we do instead?
  - socialize the cost of on-the-job training (specific skills)
  - privatize the benefit
- **The same is true for research**

Private Incentives

- A private firm wants to socialize costs; privatize benefits
- Put differently:
  - They benefit from **"Practicality" mandate**
- Perfectly rational!
- If you have a seat at the table, you get to say:
  - **Impact partnerships** = I get bargain R&D.
  - **Research ecosystems** = I get bargain R&D.
  - **Co-creation** = I get bargain R&D.
  - … sensing a pattern?

Overall

- The unicorn gets to live
- Society doesn't get to enjoy much of the unicorn it paid for

Is Behavioural Science Any Different?

No!

- Kahneman and Tversky studied optical illusions;
- Ended up devising prospect theory (among other things)
- Their work (and others') shapes a huge chunk of modern applications of behavioural science
- **Still true:** We are so bad at prediction that we should not try

Should we have known better?

Physician, Heal Thyself

- Replication crisis, anyone?
  - Shoehorning practicality gets us TED talk science
- On the behavioural primitives:
  - Risk preferences, time preferences, illusion of knowledge, …
  - We are ideally placed to understand these!
- Incentives won by Technical KO

Relevance to Australia

We are not Silicon Valley

- We are being sold a US Venture Capital model of science.
- **Our Strength:** Public Goods (Medicare, HECS, CSIRO).
- **The Risk:** Importing a "User Pays" logic into a public system.
- **More importantly, VC Myth:** Tesla / SpaceX / Palantir / etc

Practical Implications

(or the irony of taking practical implications from a talk which argues against the tyranny of practical implications)

Accountability Without Wishful Thinking

1. **Portfolio Accountability:** Assess the *portfolio*, not the project
2. **Disagreement Fund:** Allocate e.g. 20% of grants to proposals where reviewers *disagree*
3. **Golden Share:** No more gift unicorns. Public money funds the risk => public gets a share of the IP
4. **Stop Apologizing:** Curiosity is the only strategy that works in the long run.

Conclusion

The most practical thing we can do is protect the space for the impractical

Thank You

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